Haihao Lin

linhaihao24@mails.ucas.ac.cn Homepage GitHub Google Scholar ORCID

Research interests: Multimodal representation learning and reasoning; continual learning and adaptation; world modeling and embodied intelligence.

Education

University of Chinese Academy of Sciences (UCAS) Sep. 2024–Jun. 2027 (expected)
M.S. in Computer Science (Artificial Intelligence). Advisor: Prof. Xiangsheng Huang.

Jiangsu University Sep. 2020–Jun. 2024
B.E. in Computer Science and Technology. GPA rank: **1/171**.

Research Highlights

FiberTune: Preserving Action-Fiber Visual Residuals in Vision-Language-Action Fine-Tuning
First and corresponding author | Accepted at CoRL 2026
Paper | Code | Project

- Developed FiberTune to preserve transferable visual structure during action-supervised VLA fine-tuning. An action probe filters out action-predictive components; the resulting visual residual is aligned to a frozen visual teacher and regularized with an effective-rank prior, without adding inference overhead.
- Evaluated FiberTune across π0.5 and OpenVLA-OFT, the CALVIN and LIBERO benchmarks, and a real-world SO-101 task. Improved CALVIN ABC→D five-step success from 61.4% to 72.1% and SO-101 task success from 72.7% to 78.1% over task-loss-only fine-tuning; built the SO-101 data-collection and out-of-distribution evaluation pipeline using LeRobot.

Honors and Awards

- National Scholarship (twice); Jiangsu Provincial Merit Student; Jiangsu University President's Medal; Outstanding Graduate; Outstanding Undergraduate Thesis.
- Three silver medals at ICPC Asia Regional Contests; highest-ranked bronze-medal team at the 2024 ICPC Asia-East Continent Final; two First Prize awards at the Lanqiao Cup National Finals, one each in C/C++ and Python (Python: 1st in Jiangsu, 9th nationally).

Engineering Projects

ROS2 Systems Engineering for a Commercial Cleaning Robot
Mar. 2026–Aug. 2026

- Designed and implemented autonomous charging-dock return and docking, cliff detection and avoidance, and a docking simulator. Tuned parameters and stress-tested the docking system in simulation, then deployed and validated it on the robot using ROS2.
- Implemented the initial coverage-path planner and adapted an external planning algorithm. Optimized wake-word detection and speech recognition for the robot's noisy operating environment, integrated these functions into its state machine, and completed system-level debugging and testing.

RGB-D Grasping Prototype for Logistics
Aug. 2025–Jan. 2026

- Built an xArm-based RGB-D tabletop grasping prototype for logistics applications and deployed it on Jetson, including hand-eye calibration, suction and gripper integration, and Web API integration.

Property Prediction and Screening for an AI-Enabled Pharmaceutical Platform
Sep. 2023–Jun. 2024

- Processed data and reproduced existing models for the large-molecule screening component of an AI-enabled pharmaceutical platform; developed a property-prediction and screening system and contributed to service API implementation and platform integration. The resulting thesis received the university's Outstanding Undergraduate Thesis award.

Open-Source Contributions

MaaAssistantArknights Open-Source Automation Project
Jun. 2022–Apr. 2026 | GitHub 20k+ stars

- As an early key contributor to MaaAssistantArknights, developed multiple C++ core modules and WPF GUI features, and implemented and iteratively improved a task-planning algorithm for mutual-exclusion constraints.

HoshinoBot Plugin Development and Framework Extensions
Oct. 2021–Nov. 2025 | Main framework: approximately 1.5k GitHub stars

- Developed and maintained aichat and other plugins, integrating models such as DeepSeek through OpenAI-compatible APIs with per-group configuration, multi-turn context, persona management, persistent sessions, model switching, and voice replies. Built non-invasive message-trigger conflict detection tools to reduce unintended plugin responses, and contributed to other group-chat utilities.

Teaching and Leadership

Jiangsu University Programming Contests

- Served as captain of Jiangsu University's programming contest team and helped establish its training and selection system. Independently set and reviewed all warm-up and main-contest problems and wrote their editorials for the 2022 and 2023 university contests; also served as an on-site proctor and answered contestants' questions.

Lanqiao Cloud Courses – Python Course Development

- Independently developed the Python preparation course for the 17th Lanqiao Cup, including curriculum and algorithm content; contributed problems to the online judge problem bank.

Technical Skills

Programming and frameworks: Python, C++, PyTorch, LeRobot, and RLinf.
Embodied models: π0.5, OpenVLA-OFT, SmolVLA, and FastWAM.
Robotics and systems: ROS2, CMake, OpenCV, RealSense/RGB-D, xArm SDK, Jetson/Rockchip, Linux, and Docker.